

Statistical Decision Making 45-752

Models with Interactions of Variables
&
Modeling Choice

This file is named: 09_slides.pptx
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Statistical Decision Making 45-752

Models with Products of Variables: Interactions

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Managerial Policy Questions Regarding Water Conservation

In the water conservation experiment, individuals who received treatment three (comparison to their neighbors' water consumption) received the mailing regardless of whether they had previously been a high or low user of water.

Managerial policy questions:

- Did sending the mailing to low users encourage them to conserve, or encourage them to use more water?
- Was there a significant difference in response to the mailing between high and low users?

We now extend our statistical tool kit to answer this kind of question.

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Steps in our analysis:

Create an indicator variable to distinguish high and low users.

- How? We have data for pre-experiment water usage, i.e., (WATER_2006)
- Create an indicator variable equal to 1 for household who were above median in 2006 and 0 for households who were below median in 2006.

Estimate a regression to allow for the effect of the experiment to differ between those who were previously high users and those who were previously low users.

Test the difference between the preceding two estimates.

This is called “difference-in-differences” analysis and is a useful method in many applications.

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We first find the median water usage in year 2006

```
# find median water use in 2006
Water_Conservation_Data["WATER_2006"].median()
```

We find that median water usage in 2006 was 46 thousand gallons.

We next create an indicator variable equal to 1 if household used more than 46 and 0 otherwise.

The following command creates a column containing the indicator variable High and places the column in the Water Conservation data set.

```
Water_Conservation_Data["High"] = numpy.where(Water_Conservation_Data["WATER_2006"] >= 46, 1, 0)
```

Variable High equals 1 if a household was a high water user in 2006.

Variable High equals 0 if a household was a low water user in 2006.

[More detail for above command.](#)

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Select the subset of observations in the control group and treatment group 3.

```
# Select the subset of observations in the control group and treatment group 3
WatDat34 = Water_Conservation_Data.query("GROUP==3 | GROUP==4")
```

Estimate the following regression:

```
RegH_T3 = tpstats.model("cs", "SUMMER_07 ~ High + TREAT3 + I(High*TREAT3)", WatDat34)
RegH_T3.summary()
```

A product of two variables is called an interaction. Hence, **High*TREAT3** is an interaction.

As shown in the command above, an Interaction is enclosed in parentheses and preceded by an I.

On the next slide, we will work through the interpretation of each population coefficient in the above model. We will then be in a position to interpret each of the coefficients in our estimated equation.

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Let's write out the details for interpretation of the model.

Our estimated model is: $\hat{y} = b_0 + b_1\text{High} + b_2\text{Treat3} + b_3\text{High*Treat3}$

We interpret by substituting in appropriate values of High and Treat3:

High Users Receiving Treat 3: (High=1, Treat3=1): $b_0 + b_1*1 + b_2*1 + b_3*1 = b_0 + b_1 + b_2 + b_3$

High Users In Control Group: (High=1, Treat3=0): $b_0 + b_1*1 + b_2*0 + b_3*0 = b_0 + b_1$

➤ Subtracting above yields Effect of Treat3 on High Users: $(b_2 + b_3)$

Low Users Receiving Treat 3: (High=0, Treat3=1): $b_0 + b_1*0 + b_2*1 + b_3*0 = b_0 + b_2$

Low Users In Control Group: (High=0, Treat3=0): $b_0 + b_1*0 + b_2*0 + b_3*0 = b_0$

➤ Subtracting above yields Effect of Treat3 on Low Users: b_2

The difference between Treatment 3 and Control for **High Users** minus the difference between Treatment 3 and Control for **Low users** is:

$$(b_2 + b_3) - b_2 = b_3 = -2115 \text{ gallons}$$

Notice that this “difference in differences” is the coefficient of the term **High*Treat3**

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Regression Analysis of Treatment by User Type



Our regression result is shown on next slide.

Based on our derivations on the previous slide, we can interpret all of the coefficients.

The intercept tells us that:

- Low users in the control group used $b_0 = 22.756$.

The treatment caused:

- Significant water conservation by high users: $-2.708 = (b_2 + b_3)$
- Significant water conservation by low users: $-.593 = b_2$
- Significantly more water conservation by those who were high users than by those who were low users: $-2.115 = b_3$

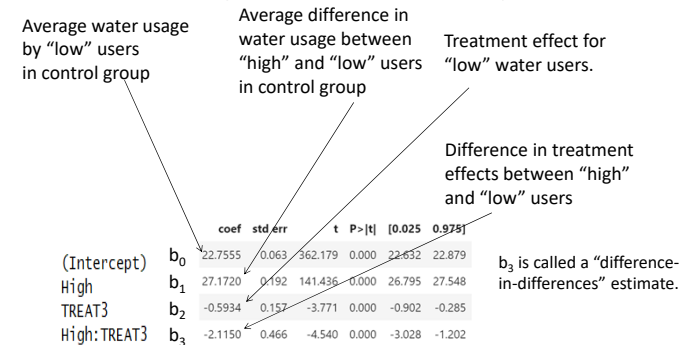
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Regression with Interaction: High*Treat



Recall that water usage is measured in thousands of gallons.



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Water Usage Summer 2007



Here is another way of displaying water usage in Summer 2007 by User Type and Experimental Assignment. Sample means are shown in aqua color.

		User Type		Difference in Differences: Difference in effect of experiment on high relative to low users
		High	Low	
Experiment Assignment	Treatment 3	47.219	22.162	
	Control	49.928	22.756	
Difference Between Treatment And Control		-2.708	-0.593	-2.115

The experiment caused high users to reduce their water consumption by 2,708 gallons.
The experiment caused low users to reduce their water consumption by 593 gallons.
High users reduced their water consumption by 2,115 gallons more than low users.

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Answers to Our Managerial Policy Questions



Did the experiment encourage low users of water to increase their consumption?

- No, it encouraged them to conserve. The effect on low users was a savings of 593 gallons per capita.

Was the effect on low users significant at the 5% level?

- Yes, p-value = .000 < .05.

Did the experiment encourage high users to conserve more than low users?

- Yes, high users conserved 2,708 gallons per capita. This is 2,115 gallons per capita more than low users.

Was the difference between high and low users statistically significant at the 5% level?

- Yes, the p-value for High*Treat3 is .0000.

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Importance of Correcting for Heteroskedasticity



If our estimates were not corrected for heteroskedasticity, we would obtain a p-value=.102 for the Coefficient of Treat3 rather than the correct p-value of .000. This shows importance of the heteroskedasticity correction.

What is the intuition? Variance in water use by high users is greater than the variance by low users.

- The following command selects data for high users in summer 2007, calculates the variance of water use for high users, and reports the variance. The variance for high users is 1,166.

```
High_Users=Water_Conservation_Data[Water_Conservation_Data['High']==1]
var_High07 = High_Users['SUMMER_07'].var()
var_High07
```

- The following does the same for low users. The variance for low users is 142.

```
Low_Users=Water_Conservation_Data[Water_Conservation_Data['High']==0]
var_Low07 = Low_Users['SUMMER_07'].var()
var_Low07
```

In summer 2007, variance of water use was much greater for high users (1,166) than low users (142).

- This is heteroskedasticity

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Terminology: Interaction



- The product of two variables is called an **interaction** of the variables.
- In our regression the interaction term, High*Treat3 is the product of two binary variables.
- In general, an interaction is a product of a pair of variables where the variables can be both continuous, both binary, or one of each.
- An interaction lets the effect of each variable depend on the value of the other variable.

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Recap Terminology: Difference in Differences



The **difference** between treatment and control for the High group is: -2708

The **difference** between treatment and control for the Low group: -593

We subtract the latter from the former to obtain the **difference in differences**:

$$-2708 - (-593) = -2115$$

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ANOVA



An analysis in which all variables are indicator variables is sometimes called Analysis of Variance (ANOVA).

- Used often to study situations in which all variables are indicator variables.
- Such situations arise in experiments (as with our water conservation case), and in many other applications (e.g., comparing effectiveness of different advertisements for a product).
- While ANOVA developed separately from regression, ANOVA is a special case of regression. Hence, you now know how to conduct a basic ANOVA analysis.

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Extending ANOVA to All Treatment Groups



Let's investigate whether there was a difference between high and low users in the other treatment groups.

We could do this separately for TREAT2 and then for TREAT3, but we can do all at once.

The following estimates all of the effects of interest in a single regression:

```
RegH_All = tprstats.model("cs", "SUMMER_07 ~ High + TREAT1 + TREAT2 + TREAT3 + I(High*TREAT1) + I(High*TREAT2) + I(High*TREAT3)", Water_Conservat)
RegH_All.summary()
```

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For Your Practice: Interpreting The Coefficients



Note that the coefficients of High, Treat3, and High*Treat3 are exactly as before.

Using the interpretation of Treat3 and High*Treat3 as a guide, interpret the coefficients of the following:

- Treat2
- High*Treat2
- Treat1
- High*Treat1

Was there a significant difference in effect of treatment 2 between high and low users?

[Answers](#)

	coef	std err	t	P> t	[0.025	0.975]
Intercept	22.7555	0.063	362.174	0.000	22.632	22.879
High	27.1720	0.192	141.434	0.000	26.795	27.548
TREAT1	-0.2702	0.170	-1.592	0.111	-0.603	0.062
TREAT2	-0.4265	0.181	-2.352	0.019	-0.782	-0.071
TREAT3	-0.5934	0.157	-3.771	0.000	-0.902	-0.285
I(High * TREAT1)	0.2598	0.534	0.487	0.627	-0.787	1.307
I(High * TREAT2)	-0.9009	0.500	-1.802	0.071	-1.881	0.079
I(High * TREAT3)	-2.1150	0.466	-4.540	0.000	-3.028	-1.202

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Answers to Questions from Previous Slide



Interpretation of coefficients:

- Coef of Treat2: The effect of treatment 2 on water consumption of Low users.
- Coef of High*Treat2: The difference in effect of treatment 2 between high and low water users.
- Coef of Treat1: The effect of treatment 1 on water consumption of Low users.
- Coef of High*Treat1: The difference in effect of treatment 1 between high and low water users.

Significance of differences in effect of treatments between high and low users:

- Treatment 2 had did not have a significantly larger effect on high than on low users with significance level .05. It was significant at the 10% level. p-value=0.071
- Treatment 1 did not have a significantly larger effect on high than on low users, p-value=0.627.
 - In fact, Treatment 1 did not have a significant effect on either high or low users?

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Joint Hypothesis Tests



The advantage of using a single regression to investigate all treatments is that we can test hypotheses involving comparisons across treatments.

Let's test the null hypothesis that the difference in treatment effects between high and low users was the same for treatment 1 and treatment 2.

```
# coefTestH(RegH_All, c("High:TREAT1=High:TREAT2"))
RegH_All.wald_test("I(High * TREAT1) = I(High * TREAT2)")
```

- We obtain p-value=.087
- The hypothesis is not rejected at the 5% level, but it is rejected at the 10% level.

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Main Effects



Consider the following model:

```
# Model with only main effects
RegMain = tprstats.model("cs", "SUMMER_07 ~ High + TREAT1 + TREAT2 + TREAT3 + High", Water_Conservation_Data)
RegMain.summary()
```

This is called a model with only **Main Effects**

- The indicator variables for treatments are included in the model.
- The indicator for High is included.
- There are no interactions (i.e., no products of indicator variables). This model assumes the difference between high and low users is the same across the four groups.
- If we had found the difference between high and low users to be the same, or nearly so, for all four groups (control and the three treatment groups), we might have opted for this model.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	22.9078	0.075	305.782	0.000	22.761	23.055
High	26.8703	0.156	172.244	0.000	26.565	27.176
TREAT1	-0.1386	0.269	-0.515	0.607	-0.667	0.389
TREAT2	-0.8776	0.250	-3.514	0.000	-1.367	-0.388
TREAT3	-1.6645	0.235	-7.071	0.000	-2.126	-1.203

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Terminology: Factors and Factor Levels



A category of indicator variables is often called a **Factor**.

The subcategories with a category are called **Factor Levels**.

Our model has two factors:

- Experimental Treatment: This factor has four factor levels: Control, Treatment 1, Treatment 2, and Treatment 3.
- Water Usage: This factor has two factor levels: Low and High.

This is useful terminology because it helps to clearly distinguish between categories and subcategories.

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Summary



We studied how to estimate and interpret models with products of variables.

Products of variables in a regression model are called "interactions".

We applied the approach to the water conservation case.

We noted that models with indicator variables and interactions of indicator variables are sometimes called Analysis of Variance (ANOVA) models.

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More Explanation Command to Create Variable High [back](#)



We used the following command to create variable High.

```
Water_Conservation_Data["High"] = numpy.where(Water_Conservation_Data["WATER_2006"] >= 46, 1, 0)
```

For each observation in the data, the command checks whether the value of WATER_2006 is greater than or equal to 46.

- If yes, then High is set equal to 1 for that observation.
- If no, then High is set equal to 0 for that observation.

This is a very efficient way to code an indicator variable in Python.

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Many Applications Have (0,1) Dependent Variable



In many applications, we want to predict an indicator variable:

- Will a person purchase our product?
- Is the transaction fraudulent?
- Will an individual have a heart attack?
- Will a borrower default?
- Will an applicant accept our offer?
- Will a customer use our company's app?
- Will a worker will commute by bus?

A model with a (0,1) dependent variable is sometimes called a **Binary Choice Model**.

- Why? Because, by definition, "binary" means there are two alternatives.

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A Prominent Application



The US National Institute of Health (NIH) created a calculator to enable an individual to calculate the probability he or she will have a heart attack during the next ten years. The goal is to increase awareness of ways individuals can decrease their heart attack risk.

When this calculator was first posted, it was accompanied by the following statement: "Your risk score was calculated using an equation."

The equation calculated and reported a **probability** of having a heart attack.

This is the type of equation and prediction analysis that we study in this note set.

The American College of Cardiology now provides a calculator here:

<https://tools.acc.org/ascvd-risk-estimator-plus/#!/calculate/estimate/>

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Estimating Models with (0,1) Dependent Variables



One approach to modeling a (0,1) dependent variable is to continue to use regression:

- When regression is used with a (0,1) left-hand-side variable, it is called a **Linear Probability Model**.
 - You should remember this terminology. When linear regression is used with a (0, 1) dependent variable, it is called a linear probability model.
 - This is just the standard linear regression model you have come to know and love:)
- Such regression models are often used to analyze data with a (0,1) dependent variable.
- The advantage of the Linear Probability Modeling approach is that the coefficients are typically easy to interpret.
- A shortcoming is that predicted probabilities from a Linear Probability Model can sometimes be negative or greater than one.

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Was There Discrimination at Airbnb



- As you probably know, Airbnb is an online marketplace where an owner of a residence can offer the residence for rental, and a prospective renter can request to rent the residence.
- Airbnb does not own properties. It charges a commission for using its service.
- A prospective renter can send an email via Airbnb to the owner of a listed property to inquire about renting the property for a specified time period.
- The owner of the property then chooses whether to respond, and whether to offer to rent the property to the person making the inquiry.
- Thus, an owner can decline by saying No or by simply not responding.
- An experiment was conducted to investigate racial discrimination on Airbnb.

The experiment was conducted by Benjamin Edelman, Michael Luca, and Dan Svirsky and the results reported by them in "Racial Discrimination in the Sharing Economy: Evidence from a Field Experiment," *American Economic Journal: Applied Economics* vol. 9, no. 2, April 2017 (pp. 1-22). You can access the full report at: <https://www.aeaweb.org/articles?id=10.1257/app.20160213>

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Design of Experiment to Investigate Discrimination



- The experiment investigated whether the response to an inquiry differed based on whether the inquiry was by a person with a distinctively white name or a distinctively black name.
- Examples of white names used in the experiment are Greg O'Brien, Laurie Ryan, Todd McCarthy.
- Examples of black names used in the experiment are Latoya Williams, Kareem Robinson, Lakisha Jones.
- Prior to the experiment, the authors did a survey asking a sample of people to review their list of names and indicate whether they think of the name as being for a black person or a white person.
- The survey results were strongly supportive of the classification of names by race.

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Implementation of The Experiment



- Messages were sent to 6,400 hosts. Each message had a name randomly chosen from the list of black and white names.
- Aside from the difference in names, the messages were identical. For example, a host might have received the following message from Laurie Ryan or Lakisha Jones.

Hi, how are you? I am interested in renting your place for a weekend: from 9/20 (Friday at night) through 9/22 (Sunday afternoon). Is there availability? Thank You!

-Name

- A record was made of whether the host accepted the request to rent the property.

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Coding and Analysis of The Data



- In the data, Accept=1 denotes that the request was accepted. Accept=0 denotes that the request was not accepted; the host declined the request or the host did not respond.
- Results shown in the table on the next page are from Table 2 of the paper referenced earlier on slide 2. In both regressions, Accept is the dependent variable.
- Two regressions are shown. In the regression in the first column, race of the guest is the independent variable. In the regression in the second column, race of the host and gender of the host are included as additional variables. Standard errors are shown in parentheses below the coefficients.
- All variables in these regressions are indicator variables. You can infer the reference category for each regression from the variables that are included in the regression.

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Do you agree with the statements below?



- Are all coefficients shown in the table statistically significant at the 5% level? Decide your answer. Then check [here](#).
- In the first regression, the average acceptance rate for white guests was 49%.
- In the first regression, the average acceptance rate for black guests was 8 percentage points below the average acceptance rate for white guests.
- This regression demonstrates that there was discrimination on Airbnb.
- In the second regression, which controls for the race and gender of the host, the average acceptance rate for black guests was 8 percentage points below the average acceptance rate for white guests.

	Dependent variable: 1 (host accepts)	
Guest is African American	-0.08 (0.02)	-0.08 (0.02)
Host is African American		0.07 (0.02)
Host is male		-0.05 (0.01)
Host has multiple listings		
Shared property		
Host has 10+ reviews		
ln(price)		
Constant	0.49 (0.01)	0.50 (0.01)
Observations	6,235	6,235
Adjusted R ²	0.006	0.009

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Impact of the Airbnb Experiment



This was clearly a valuable experiment in demonstrating that there **was** discrimination at Airbnb. This experiment was completed in 2016 and published in 2017.

The link below is to a six-year update posted in December 2022 by Airbnb. It reports the efforts that Airbnb is taking to address discrimination.

<https://news.airbnb.com/sixyearupdate/>

The link below reports on progress through year 2023.

<https://news.airbnb.com/wp-content/uploads/sites/4/2025/02/Airbnb-2024-Project-Lighthouse-Update.pdf>

Professor Tim Derdenger teaches a case on discrimination on Airbnb in his Technology Strategy course 47-872.

His course is offered in mini 3 and in mini 4.

I think you would find it a quite valuable course.

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Airbnb and Neighborhood Externalities



Airbnb encounters other challenges. Airbnb often encounters pushback from nearby homeowners who do not rent their homes.

People who rent Airbnb units are sometimes disruptive. An extreme case was The following link is to an article reporting on proposed Pittsburgh City Council legislation

- Proposed regulations would regulate and license short-term rental units like those offered by AirBnB and VRBO.
- The legislation would require the owners of short-term rentals in the city to pay for a license. It would also require an owner to obtain the approval of zoning officials if the rental is in a residential district but not attached to the owner's home. Someone responsible for property would need to be located within 25 miles of it.

<https://www.wesa.fm/politics-government/2026-02-11/airbnb-short-term-rentals-regulations>

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Senseless Loss of Life



- <https://www.post-gazette.com/news/crime-courts/2022/04/17/pittsburgh-shooting-north-side-east-allegheeny-update-public-safety/stories/202204170168>

- Two teenagers were killed and several more were injured in a shooting at a home in Pittsburgh's East Allegheny neighborhood early Sunday in what the mayor called a "senseless loss of life."

- Up to 200 people — many of them underage — were attending a large party at an Airbnb rental property near the intersection of Suismon Street and Madison Avenue on the North Side when dozens of shots rang out around 12:30 a.m., officials said.

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Answer to Question About Statistical Significance [back](#)



The smallest t-statistic is for the coefficient of Host is African American:

- $t_{\text{calc}} = .07/.02 = 3.5$

Calculate the p-value for this t-statistic:

- degrees of freedom = $6235 - 2 = 6233$

```
# Calculate p-value for AirBnB  
2 * t.cdf(-3.5, 6233)
```

We obtain .0005.

This is far below significance level $\alpha = .05$.

p-values for all other coefficients are lower still.

Hence, for each variable, we strongly reject the null hypothesis that its population coefficient is zero.

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Analyzing Recycling



Management Question: What is the relative effectiveness of alternative policies to encourage recycling of water bottles?

To address this question, the authors listed below conducted a survey of a nationally representative sample of 608 households.

Based on the respondent's state and the time period, variables for state recycling and deposit laws were matched to each respondent.

Source: W. Kip Viscusi, Joel Huber, and Jason Bell, "Promoting Recycling: Private Values, Social Norms, and Economic Incentives," American Economic Review: Papers & Proceedings 2011, 101:3, 65–70.

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Variables Obtained for Analysis of Water Bottle Recycling



	Variable	Definition
Dependent Variable	bot10p	Number of bottles out of 10 recycled
Items reported by respondents to bottle recycling survey.	green	Considers self environmentalist
	them	Neighbors would be upset if someone put recyclables in garbage
	me	Respondent would be upset if neighbors put recyclables in garbage
	scale_inc	Income in \$10,000 increments
	top_income	Top Income category (\$175,000 or more)
	msa	Lives in a metropolitan area
	n_east	Region: Northeast
	midwest	Region: Midwest
	south	Region: South
	west	Region: West
Items collected by authors of the study.	waterdep	State has deposit law covering water bottles
	nonwaterdep	State has deposit law that does not cover water bottles
	planlaws	State requires a recycling plan
	goallaws	State requires a recycling goal
	mandoplaws	State has mandatory recycling or provides an opportunity to recycle

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Further Explanation of Variables



- All variables in the preceding table from "Green" to "Mandoplaws" are indicator variables.
- In addition, "Me" and "Them" are indicator variables.
- There are a total of eight categories of indicator variables in the data:
 - Deposit laws, three subcategories: Waterdep, Nonwaterdep, No Law.
 - Recycling laws, four subcategories: Planlaws, Goallaws, Mandoplaws, No Law.
 - Respondent region of residence, four subcategories: n_east, Midwest, south, west.
 - Indicator for whether respondent lives in a metropolitan statistical area: msa
 - Indicator for whether respondent is in the top income category: top_income
 - Indicator for attitude of neighbors to recycling: "Them"
 - Indicator for respondent attitude to recycling: "Me"
 - Indicator for whether respondent is an environmentalist: "Green"
- Variables of particular managerial interest

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Recycling Law in Pittsburgh



I asked the following of Copilot:

Does the city of Pittsburgh have mandatory recycling of plastic bottles?

✓ Recycling is mandatory in Pittsburgh

The City of Pittsburgh explicitly states:

"Everyone in Pittsburgh is required to recycle!"

This mandate comes from Pennsylvania's Municipal Waste Planning, Recycling and Waste Reduction Act 101 (1988), and Pittsburgh has followed it since rolling out citywide recycling pickup in 1990.

pittsburghpa

This applies to all residents, landlords, and businesses within city limits.

✓ Plastic bottles are accepted and required to be recycled

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Coding of The Dependent Variable



Most people report either no bottles recycled or 10 out of 10 bottles recycled. Hence, I created dependent variable "Recycler" which equals:

- 0 if individual recycles less than 50% of bottles
- 1 If individual recycles more than 50% of bottles

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A Fundamental Distinction



The distinction between **percentage change** and **percentage point change** is crucial.

This is best illustrated by example: Suppose the probability of rain tonight is .25 and the probability of rain tomorrow is .30. The following are both accurate:

- The probability of rain tomorrow is **20 percent** higher than the probability of rain tonight: $100 * (.30 - .25) / .25$.
- The probability of rain tomorrow is **5 percentage points** higher than tonight: $100 * (.30 - .25)$.

Memorize this simple but important distinction!

In models with (0,1) dependent variables, predicted changes are generally percentage point changes.

The .08 difference in host response between African American and white in the Airbnb experiment is a percentage point difference.

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Bottle Recycling: Linear Probability Model



I estimated the linear probability model in the usual way by estimating a regression.

The coefficients multiplied by 100 are interpreted as percentage points because the left-hand-side variable takes on values of 0 and 1.

-An increase in income of \$10,000 increases probability of recycling by 1.1 percentage points holding other variables constant.

-Environmentalists are 12.7 percentage points more likely to recycle than non-greens.

-Strong deposit laws are predicted to increase recycling by 15.6 percentage points relative to a state with no deposit law.

-Northeast residents are 23 percentage points more likely to recycle than residents in the south.

As always: each interpretation above is the estimated effect when all other variables are held constant.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.3452	0.057	6.013	0.000	0.232	0.458
Income	0.0111	0.004	2.966	0.003	0.004	0.018
Midwest	0.0609	0.048	1.270	0.204	-0.033	0.155
Northeast	0.2308	0.059	3.912	0.000	0.115	0.347
West	0.0799	0.065	1.228	0.220	-0.048	0.208
Them	0.0010	0.039	0.026	0.979	-0.075	0.077
Me	0.1548	0.035	4.377	0.000	0.085	0.224
Green	0.1267	0.034	3.673	0.000	0.059	0.194
Goallaw	-0.0671	0.122	-0.549	0.584	-0.307	0.173
Planlaw	0.1043	0.055	1.882	0.060	-0.005	0.213
Mandrecylaw	0.1780	0.055	3.259	0.001	0.071	0.285
Nonwaterdep	-0.0045	0.057	-0.078	0.938	-0.117	0.108
Waterdep	0.1560	0.058	2.694	0.007	0.042	0.270

A Nonwaterdep law has no effect on water bottle recycling. Do you agree?

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Comments on Interpretations on Previous Slide



The interpretations on the previous slide assume the estimated relationship is causal. It seems quite plausible that these effects are causal, but possible alternatives need to be kept in mind.

Suppose strong recycling and deposit laws are adopted in states in which residents have a strong recycling ethic.

- Then the large estimated effects of Recyclaws=3 and Deposlaws=2 may reflect the recycling ethic of residents rather than the effects of the laws.

If true, why would this matter for decisions about changing policy?

- Passing the same laws in a state where residents have a weak recycling ethic might have a smaller effect or no effect at all.

How can we guard against this potential bias?

- Try to include variables to control for recycling ethic of residents.
- The authors have done this (e.g., me, them, green, indicator variables for region).

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Bottle Recycling Predicted Values

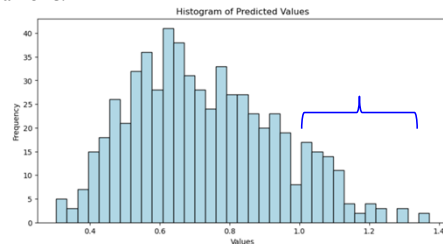


I calculated fitted values and named the variable RecylPred

This is the histogram of predicted values.

Note that **some are greater than one**.

This is a weakness of the linear probability model. Probabilities cannot be greater than one.



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Logit Models



Methods are available that assure that predicted probabilities will always lie between 0 and 1.

Two widely used approaches are called logit and probit.

Logit and Probit typically give nearly identical predictions.

Logit is a bit easier to understand, so we begin logit.

Once you understand logit, you will find it equally easy to use probit.

In logit and probit models, the coefficients are not as easy to interpret as in the linear probability model, but we will see strategies for simplifying interpretation of coefficients.

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Logistic Function



Suppose you have one right-hand-side variable x_i and the dependent variable y_i is either 0 or 1.

Let π_i be the probability that $y_i = 1$. The logit probability that $y_i = 1$ is:

$$\pi_i = \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Why use this function? Plug in any possible values of β_0 , β_1 and x_i and you will obtain a value of π_i that is between 0 and 1. You cannot get this equation to produce a probability that is less than 0 or greater than 1.

The probability that $y_i = 0$ is $(1 - \pi_i)$

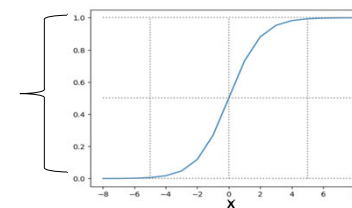
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Logistic Function



Predicted probability is always between 0 and 1.



- The logit function has the nice property that it "squashes" any value of $\beta_0 + \beta_1 x_1$ to the vertical range 0-1.
- Large negative values of x_i give a "near 0" probability while large positive values of x_i give a "near 1" probability.
- For the above graph, I set $\beta_0 = 0$ and $\beta_1 = 1$. Hence, in this illustration, the logit probability is .5 when $x = 0$.
- In applications, there can be multiple explanatory variables, and the logit function can be shifted right or left by the explanatory variables.

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Estimating Coefficients in Logit Model



The following command for our recycling data illustrates how to estimate a logit in Python. I named the estimated model RecylLogit

```
RecylLogit = tpstats.model("logit", "Recycler~Income+Midwest+Northeast+West+Them+Me+Green+Goallaw+Planlaw+Mandrecylaw+Nonwaterdep+Waterdep",
                           Bottle_Recycling_Data)
RecylLogit.summary()
```

Estimation is done using the method of Maximum Likelihood:

- Coefficients are chosen to maximize the probability of obtaining the values of the dependent variable in our sample of data.
- [The optional appendix provides more detail.](#)

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Bottle Recycling: Logit Model



The logit command on the preceding slide yields the estimates shown here.

Note that the magnitudes of **coefficients of recycling laws** show that more stringent recycling laws have larger effects on recycling.

The same is true for **stringency of deposit laws**.

Because the logit function is nonlinear, the interpretation of coefficients is not as simple as in the linear probability model.

Next slides explain how to interpret coefficients.

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.1695	0.320	-3.656	0.000	-1.796	-0.542
Income	0.0921	0.028	3.267	0.001	0.037	0.147
Midwest	0.3842	0.258	1.488	0.137	-0.122	0.890
Northeast	2.4227	0.600	4.040	0.000	1.247	3.598
West	0.0651	0.370	0.176	0.860	-0.661	0.791
Them	0.1975	0.430	0.459	0.646	-0.645	1.040
Me	1.4004	0.376	3.720	0.000	0.663	2.138
Green	0.8963	0.238	3.759	0.000	0.429	1.364
Goallaw	-0.4400	0.583	-0.755	0.450	-1.582	0.702
Planlaw	0.3705	0.281	1.317	0.188	-0.181	0.922
Mandrecylaw	1.0194	0.327	3.113	0.002	0.378	1.661
Nonwaterdep	-0.4050	0.433	-0.935	0.350	-1.254	0.444
Waterdep	1.9041	0.545	3.497	0.000	0.837	2.971

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Interpreting Logit Coefficients



In our estimated recycling model, we would like to know how the probability of being a recycler is affected by a one-unit change in each right-hand-side variable holding other variables constant.

Because the probability is a nonlinear function of the variables, we cannot interpret the logit coefficients in this way.

The effect of a one unit change in a variable depends on the values of all of the variables.

We can calculate the effect of a one-unit change in a variable from its mean when all other variables are set at their means.

These are called "marginal effects." They are interpreted as we would interpret the coefficients in a linear regression model, keeping in mind that the marginal effect of each variable is the effect of a one-unit change from the mean of the variable.

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Calculating Marginal Effects



I named our logit model RecylLogit

The following command calculates marginal effects:

```
# Calculate marginal effects
RecylLogit.marginal_effects()
```

We see that an increase in income of \$10,000 increases the probability of recycling by **1.34** percentage points.

We see that the strongest recycling law is predicted to increase the probability of recycling by **14.88** percentage points.

We see that a water deposit law is predicted to increase the probability of recycling by **27.78** percentage points.

[Derivation Underlying Above Approach](#)

Logit Marginal Effects						
Dep. Variable: Recycler						
Method: dydx						
At: overall						
	dy/dx	std err	z	P> z	[0.025	0.975]
Income	0.0134	0.004	3.388	0.001	0.006	0.021
Midwest	0.0561	0.037	1.499	0.134	-0.017	0.129
Northeast	0.3535	0.084	4.224	0.000	0.189	0.518
West	0.0095	0.054	0.176	0.860	-0.096	0.115
Them	0.0288	0.063	0.460	0.646	-0.094	0.152
Me	0.2043	0.053	3.857	0.000	0.100	0.308
Green	0.1308	0.033	3.959	0.000	0.066	0.196
Goallaw	-0.0642	0.085	-0.756	0.450	-0.231	0.102
Planlaw	0.0541	0.041	1.326	0.185	-0.026	0.134
Mandrecylaw	0.1488	0.046	3.225	0.001	0.058	0.239
Nonwaterdep	-0.0591	0.063	-0.937	0.349	-0.183	0.064
Waterdep	0.2778	0.077	3.614	0.000	0.127	0.429

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Predicted Probabilities from Logit Are Always Between 0 and 1



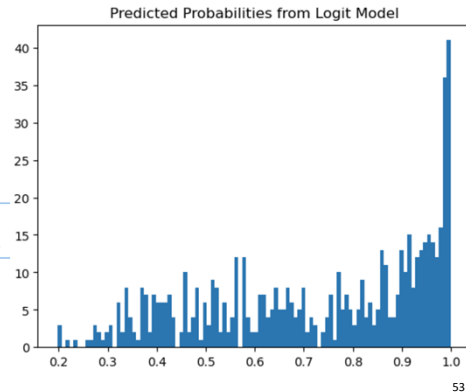
With the sample we used for estimation, we can calculate the predicted probabilities.

```
# fitted values of Logit
RecylPredLogit = RecylLogit.predict()
```

Plot a histogram of the predicted values.

```
# Histogram of Fitted Values
plt.hist(RecylPredLogit, bins=100)
plt.title("Predicted Probabilities from Logit Model")
```

All predicted probabilities lie between 0 and 1.



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Thinking About The Model



From the logit model of bottle recycling, we see that only two policies have a statistically significant effect on water bottle recycling:

mandatory recycling law

water bottle deposit law

In thinking about the model, the following issue comes to mind. The model assumes that the effects of the laws are additive.

This implies that implementing both of the above laws would reduce recycling by the sum of the two coefficients. This seems implausible.

To investigate, let's include the product of the two variables. Recall that this is called an interaction.

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Model Including Product of Policy Variables



The coefficient of the product of the two laws is negative and significant at the 5.5% level.

Suppose a state implements both laws. The combined coefficients are then:

$$(1.11 + 2.3 - 1.85) = 1.56$$

If Mandrecylaw is implemented without Waterdep, the estimated coefficient is 1.11.

If Waterdep is implemented without Mandrecylaw, the estimated coefficient is 2.3

Thus, the combined effect of both laws is less than implementing only Waterdep.

Thus, the implications of the logit model with the interaction are in close accord with the findings with the linear probability model.

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-1.1871	0.321	-3.695	0.000	-1.817	-0.557
Income	0.0947	0.028	3.343	0.001	0.039	0.150
Midwest	0.4080	0.260	1.572	0.116	-0.101	0.917
Northeast	2.4445	0.595	4.106	0.000	1.278	3.611
West	4.257e-05	0.371	0.000	1.000	-0.726	0.726
Them	0.2237	0.430	0.520	0.603	-0.619	1.066
Me	1.3988	0.376	3.715	0.000	0.661	2.137
Green	0.9078	0.239	3.791	0.000	0.438	1.377
Goalaw	-0.4432	0.584	-0.759	0.448	-1.587	0.701
Planlaw	0.3221	0.283	1.140	0.254	-0.232	0.876
Mandrecylaw	1.1095	0.334	3.324	0.001	0.455	1.764
Nonwaterdep	-0.4166	0.432	-0.964	0.335	-1.264	0.430
Waterdep	2.3000	0.609	3.779	0.000	1.107	3.493
I(Mandrecylaw * Waterdep)	-1.8538	0.965	-1.921	0.055	-3.745	0.038

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Evaluating Predictive Accuracy



When we have estimated a logit model, we would like to have a way of assessing predictive accuracy.

One way to do that is to use the predicted probabilities to predict a (0,1) outcome for each observation and then compare to the actual (0,1) outcome for each observation.

To classify predicted probabilities at 0 or 1, we choose a p_cutoff (not to be confused with a p -value). Then:

- For all observations with predicted probability $\geq p_cutoff$, we assign a value of 1.
- For all observations with predicted probability $< p_cutoff$, we assign a value of 0.

For example, scroll back to the histogram of predicted probabilities.

Suppose we choose a p_cutoff of .6.

Then all observations to the right of .6 on the horizontal axis would be classified as 1 and all to the left as zero.

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Evaluating Predictive Accuracy



It is tempting to think that a p_cutoff of .5 is a natural choice for assigning (0,1) values based on predicted probabilities.

In practice, however, there are many applications where the choice of cutoff involves tradeoffs you take into account as a manager.

We will investigate such an application later in this set of slides.

First, we will see how to calculate a classification table using `tpstats` and how to interpret the results.

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Evaluating Logit Predictive Accuracy: Classification Table



The `tpstats` command to create a logit classification table is:

```
# Calculate classification table
RecylLogit.classification_table()
```

Entries in the above command are:

- Name of logit model (RecylLogit)
- Name of left-hand-side variable in quotes (Recycler)
- Name of file containing the data (Bottle_Recycling_Data)
- p_cutoff . If none is provided (as above), p_cutoff is set equal to the average value of the dependent variable (i.e., the average of `RecylPredLogit` in this application).

The results are shown below. We did not specify a p_cutoff , so the p_cutoff is set at the default value.

- The default p_cutoff is the mean of predicted probabilities. In our data, the mean is 73%.
- With this p_cutoff , the logit equation correctly predicts **0 for 134** observations and correctly predicts **1 for 304**
- The model incorrectly predicts **0 for 140** observations and incorrectly predicts **1 for 30** observations.

Hence, total correct=438, total incorrect=170.

Summary	Actual	Predicted	Frequency
Correct	0	0	134.0
Incorrect	0	1	30.0
Incorrect	1	0	140.0
Correct	1	1	304.0

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Probit Model



The logit model employs the logistic distribution.

The probit model employs the normal distribution.

They typically give very similar results.

```
# Using probit
RecylProbit = tpstats.model("probit", "Recycler=Income+Midwest+Northeast+West+Them+Me+Green+C(Recylaw)+C(Deposlaw)", Bottle_Recycling_Data)
RecylProbit.summary()
```

We calculate marginal effects using:

```
# Probit marginal effects
RecylProbit.marginal_effects()
```

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Summary



These notes introduce analysis of models with a (0,1) dependent variable.

We illustrated use of the standard linear regression approach, which is called a linear probability model.

- The strength of the regression approach is ease of interpretation of coefficients.
- The weakness is that predicted probabilities may be negative or greater than 1.

We defined and applied the logit model.

- We did an application to bottle recycling.
- We saw that predicted probabilities are always between zero and one.
- We saw how to interpret the coefficients using marginal effects.
- We saw how to calculate and interpret a logit classification table.
- From a managerial perspective, this model would be valuable if you were doing a consulting project advising a state government on the likely effectiveness of different types recycling laws and bottle deposit laws.

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Statistical Decision Making 45-752

Modeling Choice: Application to Residential Solar Energy Systems

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Project Report from a Tepper Summer Intern

The following report is from a Tepper summer intern who had a very productive summer internship. This is the motivation for our next application.

Dear Prof. Eppler:

Sure thing. I'm happy to provide a brief description of the project I completed for my summer internship.

Background: The company sold residential solar systems. The basic flow of the business was that leads were generated by the marketing team (mostly) then a system was designed with several characteristics (size, price, current energy bill, approximate money saved etc). All systems would then go to the sales team to try and close the sale. The company had designed several thousand systems and had several hundred sales and I attempted to use the power of statistics to see if I could predict the probability of making a sale once a system was designed and before it went to the sales team.

In order to attack this project I first had to determine what variables were relevant and what was the best regression to run since linear seemed an unlikely option. After reading more about binary choice models and a few discussions with Professor Eppler, I was able to determine that a binary probit model would meet my requirements. I gave all systems that were sold a 1 and all unsold systems a 0 (accounting for time it took to make a sale) and ran the model with pretty much all of the variables I had (started with about 10). The probit model compares the variables between the sold and unsold systems to determine (with least error) how much each one effects the probability of making a sale. The beauty of the model was that once polished I could use the coefficients of each parameter in an equation using the normal distribution to determine the probability of a sale instantly after the system was designed.

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Internship Report Concludes

The exercise of narrowing down which coefficients were important was pretty much exactly what you'll be going through this mini. I first started with all the variables I had. Then using the p-values I was able to determine which variables were relevant and which were basically insignificant. One I found insignificant was the value of the house. In some cases I had to use the wald coefficient test on variables that in the original output seemed insignificant, but I felt might be correlated and more significant than the output showed. For example, two variables that turned out to be very important to the final probability of a sale were how much the customers current electricity bill was and how much money they'd save in the first year, intuitively these are related. Both of these appeared insignificant at a 90% confidence level on the original output, but when I tested the probability that both of these coefficients equaled zero (using wald) they were extremely significant (p value <0.001). I continued this exercise of running the model with several different combinations of variables, analyzing the output, running additional tests and repeating many times before I felt I got it right. But in the end I was able to recommend some very impactful changes to the company's process flow. This small project is estimated to save the company \$1 million dollars a year and increase sales conversion rate by 50%.

So I guess the moral is even though this stuff can seem overwhelming at times, stick with it, because it can be very powerful stuff.

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Analysis of Sales of Solar Panels

Our summer intern was not able to provide us with data from his company due to confidentiality. I have made up a data set that illustrates the approach used in the application.

The data contain information for 1,500 leads (prospects) developed by the marketing team.

- The data contain information about homeowners and the cost of the proposed system for each home.
- The sales team followed up to try to sell a system to all of these prospects.

Variables are as follows. All dollar values are in thousands.

Sale: equals 1 if the homeowner already purchased a system, 0 otherwise.

AnnualSave: Savings in energy cost if system is installed.

ElecBill: Current annual electricity Expenditure for the home.

AnnualCost: Annualized cost to purchase the proposed system with 20-year amortization.

DrivesHybrid: Homeowner drives a hybrid car

NearbySolar: A homeowner nearby has purchased a solar system

HomeValue: Value of the home.

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Solar Sales: Linear Probability Model



I estimated the linear probability model below.

```
# Solar sales linear probability model
LPMsolar = tpstats.model("cs", "Sale=AnnualSave+ElecBill+AnnualCost+DrivesHybrid+NearbySolar+HomeValue", Solar_Data)
LPMsolar.summary()
```

The coefficients multiplied by 100 are interpreted as percentage points because the left-hand-side variable takes on values of 0 or 1.

For example, an increase of \$1,000 in the cost of a solar system decreases the predicted probability of purchase by 10.2 percentage points holding other variables constant.

Neither **AnnualSave** nor **ElecBill** are significant. How can we investigate further?

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.1818	0.134	-1.356	0.175	-0.445	0.081
AnnualSave	0.0855	0.060	1.417	0.157	-0.033	0.204
ElecBill	0.0109	0.048	0.227	0.821	-0.083	0.105
AnnualCost	-0.1020	0.037	-2.771	0.006	-0.174	-0.030
DrivesHybrid	0.0595	0.029	2.041	0.041	0.002	0.117
NearbySolar	0.1075	0.029	3.676	0.000	0.050	0.165
HomeValue	0.0004	0.000	1.930	0.054	-7.11e-06	0.001

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Joint Test of AnnualSave and ElecBill



Let's test the hypothesis that the population coefficients of AnnualSave and ElecBill are both equal to zero:

```
# Test hypothesis that coefficients of AnnualSave and ElecBill are both equal to zero
LPMsolar.wald_test("AnnualSave=0, ElecBill=0")
```

We obtain:

p-value $\approx$.000

We very strongly reject this null hypothesis.

Kudos to our Tepper student who did this test for multicollinearity in his summer internship while developing his solar sales model!

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Investigate Removing The ElecBill Variable



Let's investigate the model without ElecBill.

```
# Solar LPM without ElecBill
LPMsolar = tpstats.model("cs", "Sale=AnnualSave+AnnualCost+DrivesHybrid+NearbySolar+HomeValue", Solar_Data)
LPMsolar.summary()
```

Note large changes in the magnitude and significance of coefficient of AnnualSave compared to previous result

- What is the intuition in this application?
- We should always think carefully about dropping a variable.
- In this case, we would expect annual savings from the solar system would be an important consideration for a homeowner, but, having controlled for annual savings, there is no compelling reason for expecting electric bill to be important. Hence, we will proceed without ElecBill. Later in these slides, as sensitivity analysis, we will investigate if our managerial implications change if this variable is included.

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-0.1797	0.133	-1.346	0.178	-0.441	0.082
AnnualSave	0.0999	0.022	4.544	0.000	0.057	0.143
AnnualCost	-0.1023	0.037	-2.771	0.006	-0.175	-0.030
DrivesHybrid	0.0597	0.029	2.058	0.040	0.003	0.117
NearbySolar	0.1074	0.029	3.674	0.000	0.050	0.165
HomeValue	0.0004	0.000	1.925	0.054	-8.36e-06	0.001

It is also quite plausible that the coefficients of **AnnualSave** and **AnnualCost** would be approximately the same magnitude and opposite sign, as they are here.

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Predicted Values from Linear Probability Model



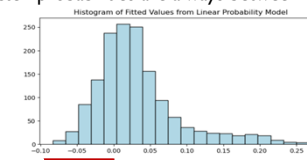
I calculated fitted values and did a histogram:

```
# Calculate fitted values and do a histogram
LPMfit = LPMsolar.predict()
plt.figure(figsize=(8, 4))
plt.hist(LPMfit, bins=20, edgecolor='black', color='lightblue')
plt.title("Histogram of Fitted Values from Linear Probability Model")
```

This is the histogram of predicted values.

Note that some predicted values are **negative**, but we know probabilities cannot be negative.

The linear probability model is helpful in giving us insights about the data. We now turn to logit to assure that our predicted probabilities are always between 0 and 1.



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Solar Sales: Logit Model



Logit estimation command is shown here:

```
# Now estimate solar sales model using Logit
logitsolar = tprstats.model("logit", "Sale~AnnualSave+ElecBill+AnnualCost+DrivesHybrid+NearbySolar+HomeValue", Solar_Data)
logitsolar.summary()
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	-12.7445	4.837	-2.635	0.008	-22.224	-3.265
AnnualSave	3.9268	0.649	6.048	0.000	2.654	5.199
AnnualCost	-4.0989	1.475	-2.778	0.005	-6.990	-1.207
DrivesHybrid	0.9644	0.413	2.333	0.020	0.154	1.775
NearbySolar	1.9421	0.398	4.884	0.000	1.163	2.722
HomeValue	0.0178	0.008	2.192	0.028	0.002	0.034

We obtain significant coefficients for all variables, and all coefficients have the anticipated algebraic signs.

We next calculate marginal effects.

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Calculating Marginal Effects



We calculate marginal effects using:

```
# Calculate average marginal effects
logitsolar.marginal_effects()
```

We obtain the following:

	dy/dx	std err	z	P> z	[0.025	0.975]
AnnualSave	0.1013	0.018	5.535	0.000	0.065	0.137
AnnualCost	-0.1057	0.039	-2.721	0.007	-0.182	-0.030
DrivesHybrid	0.0249	0.011	2.312	0.021	0.004	0.046
NearbySolar	0.0501	0.011	4.611	0.000	0.029	0.071
HomeValue	0.0005	0.000	2.161	0.031	4.27e-05	0.001

Interpretation of some coefficients:

- An increase in annual savings of \$1,000 is estimated to increase the probability of purchase by 10.1 percentage points, holding other variables constant.
- A \$1,000 increase in annualized cost of installing solar is estimated to decrease the likelihood of purchase by 10.6 percentage points, holding other variables constant.
- Holding other variables constant, the probability of purchase increases by 5 percentage points if a nearby home has had solar installed.

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Classification Table: Solar Sales



Our command in tprstats is:

```
# View classification table with default cutoff probability
logitsolar.classification_table()
```

The results are shown below. We did not specify a p_cutoff, so the p_cutoff is set at the default value.

- The default p_cutoff=.033 is the fraction of ones in our data (3.3%.)
- With this p_cutoff, the logit equation correctly predicts 0 for 1176 observations and correctly predicts 1 for 43 observations.
- The model incorrectly predicts 0 for 6 observations and incorrectly predicts 1 for 275 observations.
- Thus, the model correctly predicts 83% of the outcomes

Summary	Actual	Predicted	Frequency
Correct	0	0	1176.0
Incorrect	0	1	275.0
Incorrect	1	0	6.0
Correct	1	1	43.0

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Sensitivity Analysis



We removed ElecBill from our model. Let's investigate the difference in our logit classification if we had retained the variable.

Here is the logit model including ElecBill. I named it logitsolar2.

```
logitsolar2 = tprstats.model("logit", "Sale~AnnualSave+ElecBill+AnnualCost+DrivesHybrid+NearbySolar+HomeValue", Solar_Data)
logitsolar2.classification_table()
```

Compare to the classification table from the previous slide when we left out ElecBill, shown below right. We see that the classification accuracy is little affected with or without the ElecBill variable. Hence, we proceed without the ElecBill variable.

Summary	Actual	Predicted	Frequency
Correct	0	0	1176.0
Incorrect	0	1	275.0
Incorrect	1	0	6.0
Correct	1	1	43.0

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Classification and Decision Making



Let's investigate the effect of varying the p_cutoff

Let's try p_cutoff of .10 (result on left below).

```
# p_cutoff = 0.1
logitsolar.classification_table(0.1)
```

- With p_cutoff=.10, we predict that (83+27)=110 will purchase.
- Of these 27 actually purchased, a success rate 27/110=24.5%

Now, let's try p_cutoff of .15

```
# p_cutoff = 0.15
logitsolar.classification_table(0.15)
```

- We then predict that (46+21)=67 households will purchase.
- Of these, 21 actually purchased, a success rate of 21/67=31%

Be sure you understand the intuition for why success rate increased when we increased p_cutoff.

Summary	Actual	Predicted	Frequency
Correct	0	0	1368.0
Incorrect	0	1	83.0
Incorrect	1	0	22.0
Correct	1	1	27.0

Summary	Actual	Predicted	Frequency
Correct	0	0	1405.0
Incorrect	0	1	46.0
Incorrect	1	0	28.0
Correct	1	1	21.0

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Classification and Decision Making



It is natural to use a p_cutoff to determine which households the sales team will be asked to contact. From the previous slide, we see that:

As we increase the p_cutoff:

- The sales team will be asked to try to sell to fewer prospects.
- The sales team **success rate** is likely to increase.
- The **total number of sales** is likely to fall.

As a manager, as you consider increasing the p_cutoff, you trade off the lost profit from foregone sales against the improved success rate of the sales team (i.e., reduced cost of the sales team).

On the previous slide, increasing the p_cutoff from .10 to .15 resulted in:

- Prediction that 6 fewer systems will be sold.
- Prediction that sales team needs to contact 67 prospects instead of 110.

You must decide whether the lost profit of 6 sales is offset by the reduced time cost of the sales team to contact 110 prospects instead of 67. Put differently, is the marginal profit from 6 additional systems greater than the marginal cost of sending the sales team to 43 additional prospects?

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Prediction for Three New Observations



The following commands show how to do a logit prediction for three new prospects.

The three entries for each variable are for the three homes. For example, annual savings from a solar installation in each of the three homes are 2.5, 1.9, and 2.3.

```
Xnew = pandas.DataFrame(
    {
        "AnnualSave": [2.5, 1.9, 2.3],
        "AnnualCost": [1.8, 1.5, 1.7],
        "DrivesHybrid": [1, 0, 0],
        "NearbySolar": [0, 1, 0],
        "HomeValue": [550, 675, 495],
    }
)
```

Calculate predicted probabilities of purchase for each:

```
# Calculated predicted probability of purchase
logitsolar.predict(Xnew)
```

Our predicted probabilities of purchase for the three homes are:

```
0.613822
0.927109
0.135068
```

Which is the most promising prospect?

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Predicting Purchase Probabilities for New Prospects



We use the term "prospect" to denote a home that does not have solar but is a potential candidate for solar installation. These prospects have not been contacted by the sales team. We investigate which are the most promising.

The marketing team has obtained information for 1,000 prospects. The data are in file: Solar_New_Proprospects.xlsx

Import the file:

```
Solar_New_Proprospects = pandas.read_excel("Solar_New_Proprospects.xlsx")
PredNew = logitsolar.predict(Solar_New_Proprospects)
```

Which of these have the highest predicted probability of purchase?

The commands below rank the prospects, drop an unwanted variable, and write the results to a csv file.

```
RankNew = logitsolar.predict_and_rank(Solar_New_Proprospects)
# The above command puts a column of 1's named Intercept in the file. The next line drops that column.
RankNew = RankNew.drop(columns=['Intercept'])
RankNew.to_csv("Ranked_New_Proprospects.csv")
```

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Ranking The New Prospects



View a histogram of predicted probabilities (See next slide)

```
plt.hist(PredNew, bins=30, edgecolor='black', color='lightblue')
```

- Plot predicted probabilities against rank (See plot next slide).

```
plt.plot(RankNew["ProspectRank"], RankNew["PredictionNew"])
plt.grid(color='gray', linestyle='--', linewidth=0.5)
plt.title('Prospects Ranked by Predicted Purchase Probabilities')
plt.xlabel('New Prospect Rank')
plt.ylabel('New Prospect Predicted Purchase Probabilities')
```

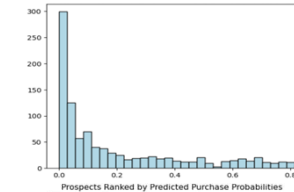
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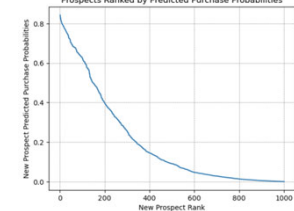
Displays of Predicted Probabilities



Histogram of predicted purchase probabilities.



Predicted purchase probabilities ranked from highest to lowest.



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Providing Information to The Sales Team



We can write the information about the best prospects to an Excel csv file named "Ranked_New_Pro Prospects" as follows:

```
RankNew = logitsolar.predict_and_rank(Solar_New_Pro Prospects)
RankNew = RankNew.drop(columns=['Intercept'])
RankNew.to_csv("Ranked_New_Pro Prospects.csv")
```

The above command, writes the csv file to your working directory.

This file shows new prospects from highest to lowest rank.

In keeping with Python convention, customers are numbered 0 to 999.

- Open the Solar_New_Pro Prospects file in Excel; and you will see that the top-ranked prospect is customer number 980, with a predicted purchase probability of 84.3%
- You can then use this file and your chosen `p_cutoff` to decide which households the sales team will contact.

A natural strategy is to have the sales team approach the highest ranked prospects first and work their way down the list until they reach the point (`p_cutoff`) where you judge that the odds of success are not large enough to justify the additional cost of contacting prospects.

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Be Sure You See the Big Picture



We obtained data for 1,500 prospects that had been approached in the past about buying a solar system.

- We estimated a logit model (logitsolar) using these data.

The marketing team provided us data on 1,000 new prospects who had not yet been approached.

- We used our model (logitsolar) to predict success with the new prospects, and we ranked the new prospects.
- We added our predicted probabilities and rankings to the file given to us by the marketing team, and we used a `p_cutoff` to determine which prospects the sales team would contact.

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We developed and applied a logit model to solve a managerial decision making problem.

- Ranking prospects for the marketing team for a solar energy company.
- As our summer intern described, this type of analysis increased the sales conversion rate for his company by 50% and increased profit by approximately one million dollars!

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We studied:

Models with a binary dependent variable.

- Estimation of Linear Probability and Logit Models
- Interpretation of Coefficients
- Prediction of Probabilities
- Creation of A Classification Table Using Predicted Probabilities

Managerial Applications

- Evaluating effectiveness of recycling and deposit laws
- Ranking Solar Sales Prospects

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The first slide that follows explains derivation of marginal effects in the logit model.

The remaining appendices are designed to explain Maximum Likelihood Estimation

A simple coin flip example is presented, and the Likelihood Function is derived for this example

Then the Logit Likelihood function is derived.

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- I glued a US quarter and a US penny together.
- George Washington's image is on one side; Abraham Lincoln's is on the other.
- If I flip my new "coin," what is the probability that Washington faces up?
- I don't know; I never will.
- Let's name this unknown and unknowable population probability π .
- Suppose we flip the coin 100 times and get "Washington" 58 times.
- Based on the data, what estimate would you give for π ?
- The intuitively obvious response is 58/100.
- This is the maximum likelihood estimator of π ; it is the value of π "most likely" to have generated our sample of 58 Washington's and 42 Lincoln's.
- Maximum Likelihood Estimation (MLE) is the most intuitive of all estimation procedures.
- We choose the coefficients most likely to generate the sample of data that we obtained.
- In this appendix, I use the coin flipping example to illustrate MLE.
- I then summarize the general approach and generalize our example to study choice probabilities.

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Example : Let π be the probability that a flip of the coin yields a head, and the probability of a tail. Flip the coin n times.

- Let y_i be the outcome on coin flip i : $y_i=1$ denotes a head and $y_i=0$ denotes a tail.

- What is the probability density for y_i ? It is the Bernoulli:

$$f(y_i) = \pi^{y_i} (1 - \pi)^{(1-y_i)}$$

- This distribution tells us the probabilities of the outcomes:
- The probability of a head is obtained by substituting $y_i=1$:

$$f(1) = \pi^1 (1 - \pi)^{(1-1)} = \pi$$

- The probability of a tail is obtained by substituting $y_i=0$:

$$f(0) = \pi^0 (1 - \pi)^{(1-0)} = 1 - \pi$$

- What is the probability distribution of our sample of n coin flips?
- The next pages provides analytic details for this example.

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The Likelihood Function

Assuming the coin flips are independent, the joint distribution is the product of the individual probabilities.

This is called the Likelihood Function, so we will denote it :

$$1. \quad L(y_1, y_2, \dots, y_n) = f(y_1)f(y_2)\dots f(y_n)$$

Substituting our Bernoulli distribution from the previous page for the $f(y_i)$:

$$2. \quad L(y_1, y_2, \dots, y_n) = \pi^{y_1} (1 - \pi)^{(1-y_1)} \cdot \pi^{y_2} (1 - \pi)^{(1-y_2)} \dots \pi^{y_n} (1 - \pi)^{(1-y_n)}$$

Collecting terms, we obtain:

$$3. \quad L(y_1, y_2, \dots, y_n) = \pi^{\sum y_i} (1 - \pi)^{(n - \sum y_i)}$$

This is the Likelihood Function for our n coin flips. It is the joint probability of obtaining the n coin flips that we obtained.

What value of π is most likely to have generated our sample of coin flips?

It is the value that maximizes the probability of getting our sample.

The Maximum Likelihood Estimator of π is the value that maximizes (3).

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Maximizing The Likelihood Function

It is convenient to work in logarithms. We always use logarithms when doing maximum likelihood estimation. The logarithm of our likelihood function is:

$$4. \quad \ln L = \sum y_i \ln \pi + (n - \sum y_i) \ln(1 - \pi)$$

To obtain the maximum, we differentiate with respect to π :

$$5. \quad \frac{\partial \ln L}{\partial \pi} = \frac{\sum y_i}{\pi} - \frac{(n - \sum y_i)}{1 - \pi}$$

The maximum is obtained at the value of π that sets the preceding to zero.

We will denote the value that maximizes the likelihood function as p :

$$6. \quad \frac{\sum y_i}{p} - \frac{(n - \sum y_i)}{1 - p} = 0$$

Solving the preceding, we obtain:

$$7. \quad p = \frac{\sum y_i}{n}$$

This is exactly as we would expect. The MLE of π is the fraction of coin flips that are heads. In our example, the maximum likelihood estimator is $p=58/100$.

This completes derivation for this example.

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The logistic function is: $\pi = \left(\frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \right)$

The derivative of the logistic function

with respect to x is:

$$\frac{\partial \pi}{\partial x} = \beta_1 * \left(\frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \right)$$

The margins command calculates the value of the derivative for each data point.

The average of these derivatives, the average marginal effect, is then reported by the margins command.

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